

AI Requirements Worksheet

#	Requirement (as written)	Source (role/stakeholders)	Conflict	Ambiguity	Completeness	Consistency	Testability	Evolvability	Notes/Resolution
1	The system shall maintain a record of every piece of rentable gear (about 150 items), including its type (tent, kayak, bike, etc.).	Inventory	PASS	PASS	PASS	PASS	PASS	PASS	
2	The system shall show each item's current status, at minimum whether it is available or rented out.	Inventory	PASS	PASS	PASS	FAIL	PASS	PASS	Resolution: reword “item” to “piece of gear”
3	The system shall record which customer is renting which item(s).	Rentals	PASS	PASS	PASS	FAIL	PASS	PASS	Resolution: reword “item(s)” to “piece(s) of gear”
4	The system shall record a due-back date and time for each rental.	Rentals	PASS	PASS	PASS	PASS	PASS	PASS	
5	The system shall record when an item is returned.	Rentals	PASS	PASS	PASS	PASS	PASS	PASS	
6	The system shall keep a rental history for each item, so staff can find who had it last and who had it at any past point.	Accountability	PASS	PASS	PASS	PASS	PASS	PASS	
7	The system shall let staff record damage when an item is returned, linked to the rental during which it happened.	Accountability	PASS	PASS	PASS	PASS	PASS	PASS	
8	The system shall identify items that are overdue or have not been returned	Accountability	PASS	PASS	PASS	PASS	PASS	PASS	
9	The system shall support about 150 items and about 50 rentals per weekend day without slowing down.	Scale and Simplicity	PASS	FAIL	PASS	FAIL	PASS	PASS	Resolution: “without slowing down” is ambiguous, and “items of gear” should be used instead of “items”. Rewrite: The system shall support about 150 pieces of gear and about 50 rentals per weekend day while maintaining at least 97% of its baseline speed when loading and updating rentals and pieces of gear.
10	Staff shall be able to learn and use the system with minimal training.	Scale and Simplicity	PASS	FAIL	PASS	PASS	FAIL	PASS	Resolution: “with minimal training” is unclear. Rewrite: Staff shall be able to learn and use the system with 15 minutes or less of training.
11	Each item shall have a unique ID, such as a tag or sticker number, so identical items (e.g., three of the same tent) can be told apart.	Gear identification and condition	PASS	PASS	PASS	PASS	PASS	PASS	
12	Staff shall record an item's condition at checkout as well as at return. Without a checkout baseline, damage can't be reliably attributed to a specific renter.	Gear identification and condition	PASS	PASS	PASS	PASS	PASS	PASS	
13	Items shall have an "out of service" or "in repair" status so damaged gear isn't rented again by mistake.	Gear identification and condition	PASS	PASS	PASS	PASS	PASS	PASS	
14	The system shall let staff retire items that are lost or permanently damaged, while keeping their history.	Gear identification and condition	PASS	PASS	PASS	PASS	PASS	PASS	
15	The system shall store customer contact details (name, phone,	Customers	PASS	PASS	PASS	PASS	PASS	PASS	

	possibly ID or license number) so staff can follow up on overdue or damaged gear.								
16	A single rental shall be able to include multiple items. For example, a family might rent two kayaks, paddles, and life jackets together.	Customers	PASS	PASS	PASS	PASS	PASS	PASS	
17	Returning customers can be looked up rather than re-entered.	Customers	PASS	FAIL	PASS	PASS	PASS	PASS	Resolution: requirement is too vague “The system shall be able to quickly look up a frequently returning client.”
18	Staff can see a list of all items currently out and when each is due back.	Operations	PASS	PASS	PASS	PASS	PASS	PASS	
19	Staff can check whether an item type is available for a given date, if the shop takes reservations.	Operations	PASS	PASS	FAIL	FAIL	PASS	PASS	Resolution: requirement is vague and oddly worded “The system shall be able to check when out of stock items will become available again”
20	The system shall support partial returns, where some items in a rental come back earlier than others.	Operations	PASS	FAIL	PASS	PASS	PASS	PASS	Resolution: requirement is too vague “The system shall support early item returns for certain singular rentals that include multiple items.”
21	The system may record rental fees, deposits, and damage or late charges.	Money	PASS	FAIL	FAIL	PASS	FAIL	PASS	Resolution: requirement is too vague “The system shall record a client’s rental fees, current deposits, damaged item history, and past late fees”
22	Checking gear out or in should take no more than a minute or two, so lines don't form on busy weekends.	Non-functional	PASS	PASS	PASS	PASS	PASS	PASS	
23	Customer data shall be stored securely and viewable only by staff.	Non-functional	PASS	PASS	PASS	PASS	PASS	PASS	
24	Data shall be backed up so records survive a device failure.	Non-functional	PASS	PASS	PASS	PASS	PASS	PASS	
25	The system shall remain usable if the internet connection is weak. Trailheads are often remote.	Non-functional	PASS	PASS	FAIL	FAIL	FAIL	PASS	Resolution: requirement is too specific to test, is also unnecessary as a system requirement “The system shall be usable, given that the system is currently connected to the internet.”